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When Every Second Counts: Analysis of Fire Response Time Compliance

I. Business Need

When a 911 call is placed, the margin for error is measured in seconds. The American Heart Association states that brain death begins within four to six minutes of cardiac arrest, with survival odds falling seven to ten percent for every minute without defibrillation (Zijlstra et al., 2018). Research also indicates that modern residential fires can reach flashover, the point at which every surface in a room ignites simultaneously, in under five minutes (Kerber, 2012). To standardize response across fire departments, the National Fire Protection Association developed NFPA 1710 which requires turnout within 80 seconds for fire calls and 60 seconds for EMS calls, with first responders arriving on scene within four minutes on at least 90% of all incidents (NFPA, 2020). These thresholds are not arbitrary but rather grounded in decades of resuscitation research and fire behavior research that links arrival time to survivability. Despite the national standard, an analysis of the Milwaukee Fire Department's 2023 incident records reveals a significant gap in compliance. Of the 848,242 responses to calls examined, 23.7% failed to meet NFPA 1710 combined turnout and travel time standard of 320 seconds which is well below the required 90% target. Upon further investigation, miss rates are distributed consistently across days of the week and months of the year which suggests an operational challenge rather than isolated scheduling issues. If leadership in Milwaukee County can predict which incidents are most likely to miss the standard, resources can be strategically pre-positioned, and high-risk neighborhoods can receive increased attention. Data mining techniques applied to 2023 Milwaukee Fire Department records were used to identify whether a given emergency response will meet or miss the NFPA 1710 standard and to determine what factors drive that outcome.

II. Statistical Methodology

a. Data Management

Four datasets sourced from the Milwaukee Fire Department's 2023 public records were merged into a single unit-level analytical file, combining fire incident details, unit status records, and a ZIP code-level daily summary of EMS demand. Data management steps included restricting records to the City of Milwaukee, removing invalid timestamps, eliminating negative time values, and capping total response times at two hours. Four time-based variables were engineered from cleaned timestamps — dispatch time, turnout time, travel time, and total response time — alongside temporal features including hour of day, day of week, and month. The binary outcome variable, Missed_Standard, was defined as one when combined turnout and travel time exceeded 320 seconds, directly reflecting the NFPA 1710 standard. After all cleaning steps the final dataset contained 848,242 unit-level records, of which 23.7 percent missed the standard.

b. Performance Evaluation Framework

The data set was partitioned into training and validation sets using a stratified 60/40 split to preserve the original distribution of 23.7% missed standards across both sets. This was done first to set a benchmark for the following analyses. Confusion matrices, sensitivity, specificity, lift chars and ROC curves were applied within the classification tree and random forest analysis. An adjusted classification cutoff of 0.237 was made to all models to reflect the true class distribution with the model tuned to catch as many missed responses as possible because every missed response that is undetected represents an emergency where help did not arrive in time. Performance evaluation was not applied to the logistic regression model as it was used to identify significant predictors and relationships rather than the primary classification model.

c. Logistic Regression

A logistic regression model was estimated to identify which factors significantly predict a missed standard and in what direction, with EMS_High_Severity excluded as a predictor due to multicollinearity identified previously.

d. Classification Tree

A classification tree was used to create a visualization in which an individual can identify the operational factors that separate incidents that meet the standards from those that did not. A full tree was used to identify all the possible splits and determine where pruning would be most appropriate to create a digestible visualization using Hour, Day of the Week, Month, ZIP Code, EMS Total Calls, EMS High Severity and EMS Escalated as predictors. The tree was then pruned using a complexity parameter of 0.001. Full performance evaluation was then applied to assess classification accuracy, sensitivity and specificity.

e. Random Forest

A Random Forest was employed to improve upon the classification by combining many decision trees to determine the final classification, reducing the instability and overfitting associated with a single tree. Variable importance scores were extracted to identify which operational factors most strongly drive missed standards across the full dataset. The ranger package was used in place of the traditional randomForest package to use multiple CPU cores, given the computational demands of an 848,242 record dataset. Full performance evaluation was applied to assess classification accuracy, sensitivity, and specificity.

III. Results

a. Exploratory Data Analysis

Exploratory analysis of the 848,242 incident records revealed that 23.7% of responses failed to mee the NFPA 1710 combined turnout and travel time standard of 320 seconds. Since the miss rates were consistent across all days of the week and months of the year it suggested that there was an operational challenge. Geographic analysis also revealed meaningful variation in miss rates across ZIP codes with several neighborhoods in the northwestern and southwest areas of Milwaukee showing disproportionately high miss rates (Figure 1). Peak call volume was observed in July during evening hours, though elevated demand alone did not fully explain the compliance gap.

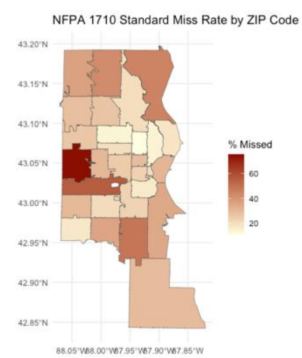

Figure 1. Standard Miss Rate by ZIP Code

b. Logistic Regression

Logistic regression identified several operation factors that significantly predict a missed standard. Turnout time was the strongest predictor, followed by hour of day and dispatch time, indicating that crew readiness and dispatcher speed are the primary operational drivers of compliance failures. Higher EMS demand in a ZIP code in each day also significantly increased the probability of missing the standard which suggests system congestion effect. Several ZIP codes such as 53225, 53217, 53214, and 53223 exhibited significantly higher odds of missing the standard even after controlling for all other variables, pointing to geographic resource gaps (Table 1).

Table 1. Logistic Regression Model Significant Predictors of Missed Standard

Variable	Coefficient
Turnout Time	0.014
Hour	-0.029
Dispatch Time	0.0007
EMS Total Calls	0.0049
EMS Escalated	-0.0402
ZIP Code 53226	2.636
ZIP Code 53217	1.683*
ZIP Code 53221	1.61*
ZIP Code 53214	1.544*
ZIP Code 53223	1.355*

*All variables shown are statistically significant at $p < 0.001$ unless otherwise noted. ($p < 0.005$) **

c. Classification Tree

The pruned classification tree produced five splits using only Hour and ZIP Code as splitting variables, suggesting that time of day and geographic location are the two most discriminating factors in separating incidents that met the standard from those that did not (Figure 2). The simplicity of the pruned tree is both a strength and a limitation where the if-then rules are highly interpretable and actionable for fire department leadership, the model sacrifices predictive complexity in favor of readability. At the adjusted cutoff of 0.237 the model achieved an accuracy of 71.66%, a sensitivity of 33.91%, a specificity of 83.35%a Kappa of 0.180, and an AUC of 0.611, indicating weak discriminatory power overall. The low sensitivity is the most concerning finding because the model correctly identified only 33.91% of incidents that actually missed the standard, meaning the majority of missed responses were misclassified as on time. While the classification tree serves as a useful interpretable baseline, these results signal the need for a more powerful modeling approach capable of capturing the complexity of the underlying patterns in the data.

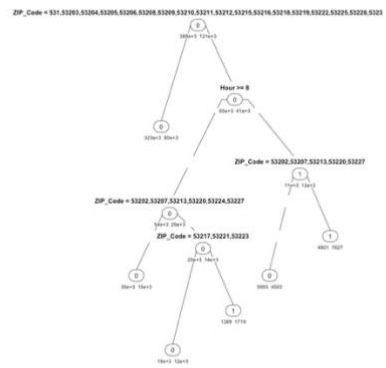

Figure 2. Standard Missed Classification Tree

d. Random Forest

The random forest substantially outperformed the classification tree across all evaluation metrics, achieving an accuracy of 90.76%, sensitivity of 94.25%, specificity of 89.69%, a Kappa of 0.766 and an AUC of 0.978. The ROC curve measures the model's ability to distinguish between incidents that missed the standard and those that did not across all possible classification thresholds, with an AUC of 0.978 indicating that the random forest correctly ranks a missed response above an on-time response 97.8% of the time (Figure 3).The high sensitivity is particularly meaningful in this context in which the model correctly identified 94.25% of all incidents that missed the standard, meaning very few missed responses went undetected. Variable importance scores identified ZIP Code and Hour as the strongest predictors, followed closely by EMS Total Calls, EMS High Severity, Month, Day of the Week and EMS escalated, confirming that geography, time of day and system demand are the primary operational drivers of missed standards. The geographic finding is also especially notable as ZIP Code ranking as the single most important variable suggests that certain neighborhoods face structural response time disadvantages that persist even after controlling for demand and temporal factors. A Decile-Wise lift chart further demonstrated the model's discriminatory power in which the top ten percent of scored incidents captured 42.1% of all missed standards, representing a lift of 4.21 times better over random selection, while the bottom 60% of incidents contained virtually no missed standards at all (Figure 4).

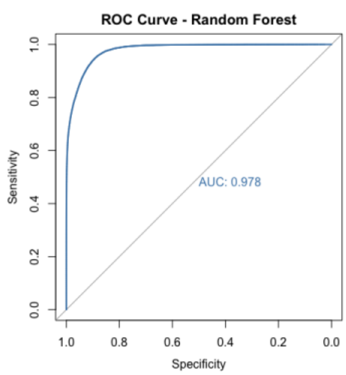

Figure 3. ROC Curve Random Forest

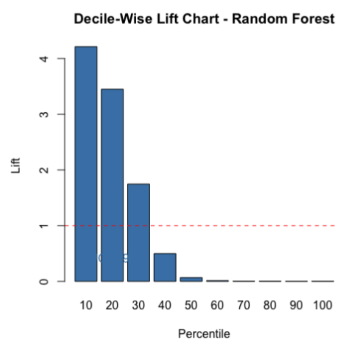

Figure 4. Decile Wise Lift Chart of Random Forest

IV. Alternative approaches

Principal Components Analysis was explored as a dimension reduction technique to eliminate multicollinearity among predictors prior to logistic regression, however a correlation matrix revealed only one meaningful correlation among the six predictors examined and retaining sufficient variance would have required all six components, defeating the purpose of dimension reduction. Boosting was considered as an alternative ensemble method that builds trees sequentially to correct prior errors, however unlike boosting the random forest builds trees independently in parallel, reducing variance rather than bias, which proved more appropriate for a large and noisy dataset of 848,242 incidents, and the resulting AUC of 0.978 left minimal room for meaningful performance improvement. A regression tree was considered but deemed inappropriate as the outcome variable, whether a response missed the NFPA 1710 standard, is binary in nature, making a classification approach the correct analytical framework.

V. Conclusion

This analysis set out to determine whether the operational factors driving Milwaukee Fire Department response time compliance failures could be identified and classified using data mining techniques applied to 2023 incident records. The random forest emerged as the strongest classification model, achieving 90.76% accuracy, 94.25% sensitivity, and an AUC of 0.978, correctly identifying the vast majority of incidents that missed the NFPA 1710 standard. With 23.7% of 848,242 unit-level responses failing to meet the 320 second combined turnout and travel time standard, Milwaukee falls 13.7 percentage points below the 90% compliance target set by NFPA 1710. ZIP Code ranked as the single most important predictor variable, suggesting that certain Milwaukee neighborhoods face structural response time disadvantages that persist independent of demand and time of day. Turnout time and hour of day were the next strongest drivers, indicating that crew readiness and time of day are the most actionable operational levers available to department leadership. The logistic regression further confirmed that higher EMS demand within a ZIP code on a given day increases the probability of a missed response, pointing to a system congestion effect that compounds geographic disadvantage. These findings collectively suggest that targeted resource reallocation to high-risk ZIP codes, improvements in crew turnout procedures, and demand-aligned staffing during peak hours represent the highest leverage opportunities for closing Milwaukee's compliance gap. Data mining techniques applied to routine incident records offer leadership a powerful and accessible framework for identifying where and when compliance failures are most likely to occur and directing resources accordingly.

Appendix A.

Figures.

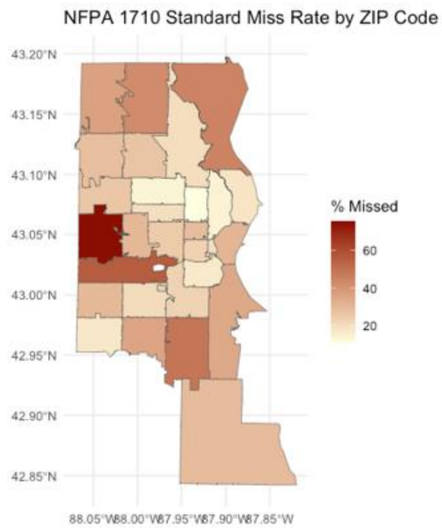

Figure 4. Standard Miss Rate by ZIP Code

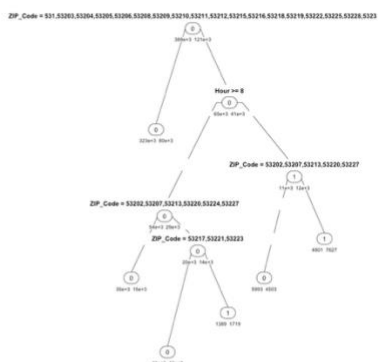

Figure 3. Standard Missed Classification Tree

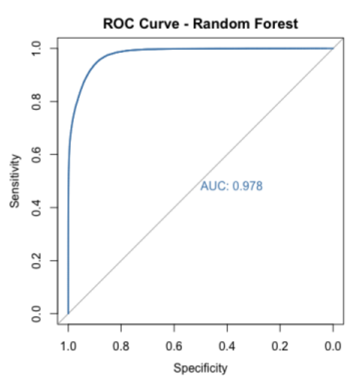

Figure 3. ROC Curve Random Forest

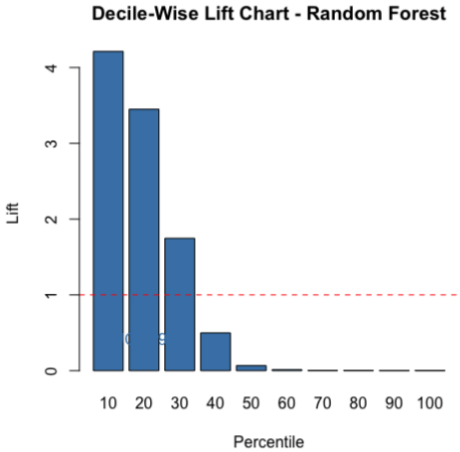

Figure 4. Decile Wise Lift Chart of Random Forest

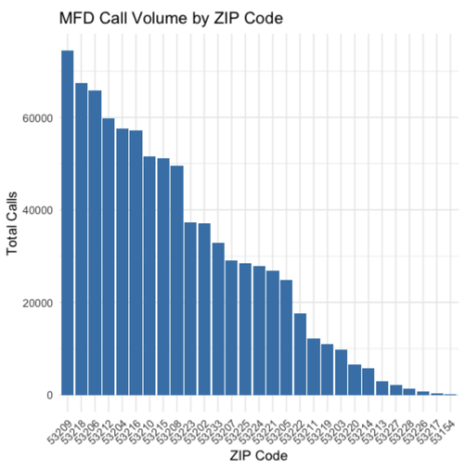

Figure 5. Call Volume by ZIP Code

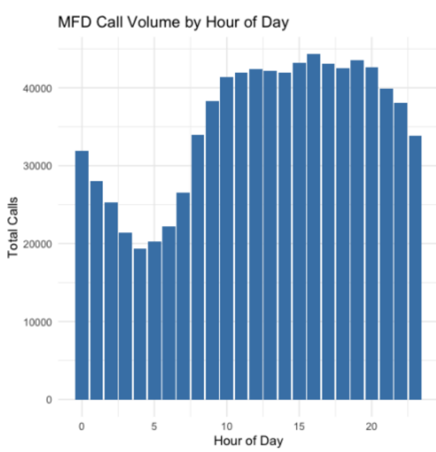

Figure 6. Call Volume by Hour

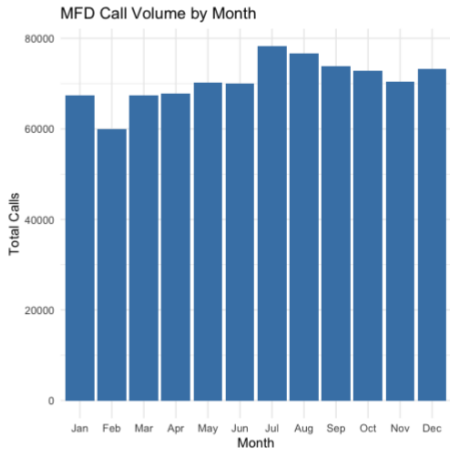

Figure 7. Call Volume by Month

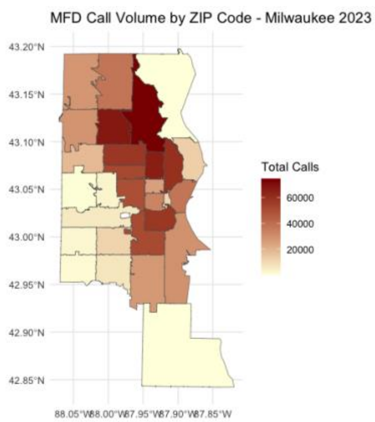

Figure 9. Call Volume by ZIP Code

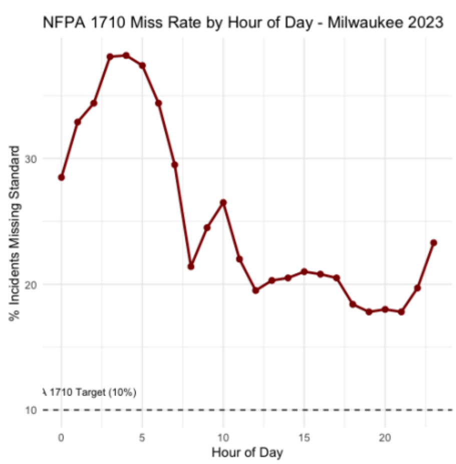

Figure 8. Miss Rate by Hour

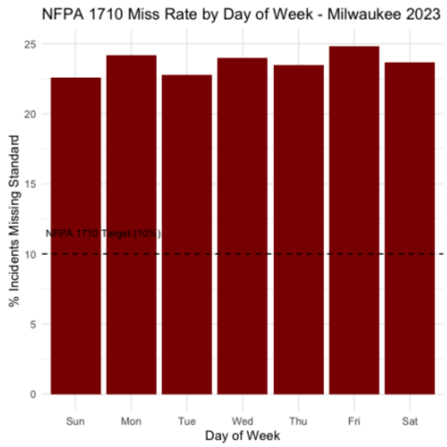

Figure 10. Miss Rate by Day

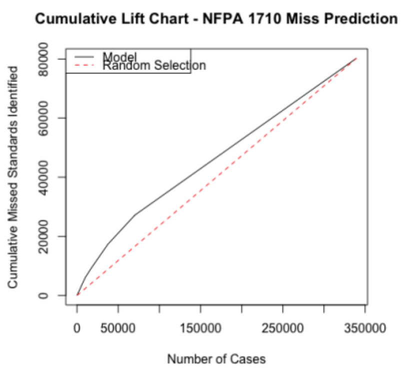

Figure 12. Cumulative Lift Chart for Classification Tree

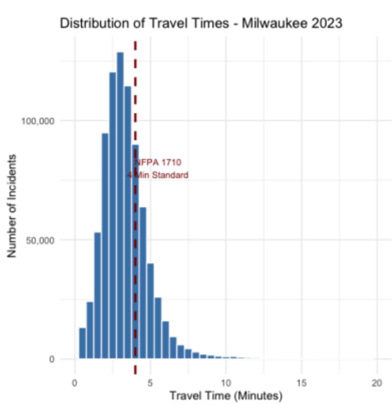

Figure 11. Distribution of Travel Times

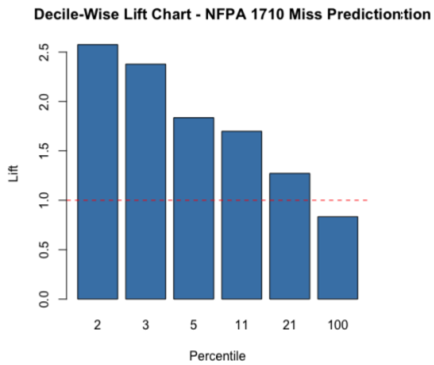

Figure 13. Decile-Wise Lift Chart of Classification Tree

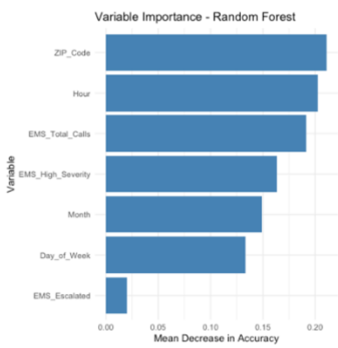

Figure 15. Variable Importance - Random Forest

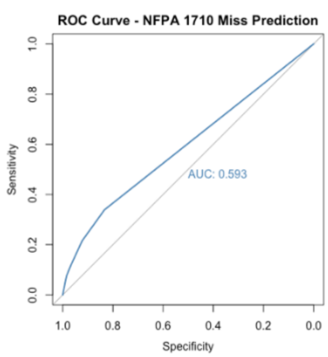

Figure 14. ROC Curve of Classification Tree

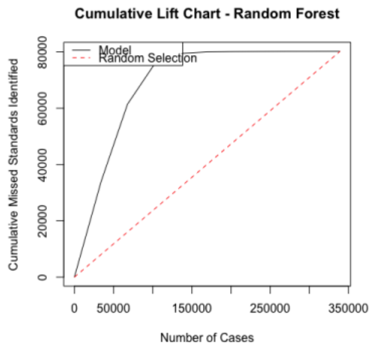

Figure 16. Cumulative Lift Chart of Random Forest

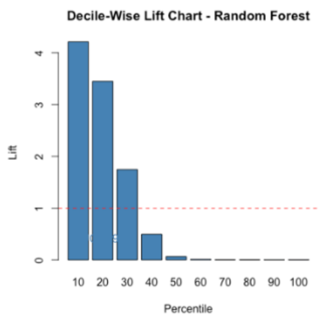

Figure 17. Decile-Wise Lift Chart - Random Forest

Appendix B.

Tables.

Table 1. Logistic Regression Model Significant Predictors of Missed Standard

Variable	Coefficient
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ZIP Code 53214	1.544*
ZIP Code 53223	1.355*

*All variables shown are statistically significant at $p < 0.001$ unless otherwise noted. ($p < 0.005$) **

Table 2. Confusion Matrix — Classification Tree (Adjusted Cutoff 0.237)

	Reference (Actual)	
Prediction	**Met (0)**	**Missed (1)**
Met (0)	215,953	53,012
Missed (1)	43,130	27,201

Accuracy: 71.66% | Sensitivity: 33.91% | Specificity: 83.35% | Kappa: 0.180

Table 3. Confusion Matrix — Random Forest (Adjusted Cutoff 0.237)

	Reference (Actual)	
Prediction	**Met (0)**	**Missed (1)**
Met (0)	232,112	4,558
Missed (1)	26,971	75,655

Accuracy: 90.71% | Sensitivity: 94.32% | Specificity: 89.59% | Kappa: 0.765

Appendix C.

References.

Zijlstra, J. A., Stieglis, R., Riedijk, F., Smeekes, M., van der Worp, W. E., & Koster, R. W. (2018). Local lay rescuers with AEDs, alerted by text messages, contribute to early defibrillation in a Dutch out-of-hospital cardiac arrest dispatch system. Circulation, 136(1). https://doi.org/10.1161/circulationaha.117.029067

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